

Interacting Multipoles of Rank 4 and 4:

[illegible]

Simplify deltas

[illegible]

Simplify R

[illegible]

Simplify Multipoles

[illegible]

Exploit Traceless Conditions

[illegible]

Reduce Indices

[illegible]

Add up

$$T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = R_z^{-17} \cdot -\frac{1}{135135} \cdot \left[\begin{aligned} &-540540 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{zzz\delta} \\ &-540540 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{zzz\gamma} \\ &-540540 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{zzz\beta} \\ &-540540 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{zzz\alpha} \\ &+20790 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\eta\eta} \\ &+41580 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\zeta\zeta} \\ &+62370 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\ &+20790 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\eta\eta} \\ &+41580 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\ &+62370 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\ &+10395 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{zz\eta\eta} \\ &+124740 \cdot R_z^8 \cdot \Phi_{zz\gamma\delta} \cdot \Phi_{zz\gamma\delta} \\ &+20790 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\zeta\zeta} \\ &+31185 \cdot R_z^8 \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\epsilon\epsilon} \\ &+20790 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\eta\eta} \\ &+41580 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\zeta\zeta} \\ &+62370 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\ &+20790 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\eta\eta} \\ &+124740 \cdot R_z^8 \cdot \Phi_{zz\beta\delta} \cdot \Phi_{zz\beta\delta} \\ &+41580 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\zeta\zeta} \\ &+62370 \cdot R_z^8 \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\epsilon\epsilon} \\ &+124740 \cdot R_z^8 \cdot \Phi_{zz\beta\gamma} \cdot \Phi_{zz\beta\gamma} \\ &+41580 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\beta\beta\delta} \\ &+41580 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\beta\beta\gamma} \\ &+41580 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\gamma\gamma} \\ &+20790 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\eta\eta} \\ &+41580 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\zeta\zeta} \\ &+62370 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ &+31185 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\eta\eta} \\ &+124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\delta} \cdot \Phi_{zz\alpha\delta} \\ &+62370 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\zeta\zeta} \\ &+93555 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\ &+124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\gamma} \cdot \Phi_{zz\alpha\gamma} \\ &+83160 \cdot R_z^8 \cdot \Phi_{zzz\delta} \cdot \Phi_{z\alpha\alpha\delta} \\ &+83160 \cdot R_z^8 \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\alpha\alpha\gamma} \\ &+41580 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\ &+124740 \cdot R_z^8 \cdot \Phi_{zz\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\ &+83160 \cdot R_z^8 \cdot \Phi_{zzz\beta} \cdot \Phi_{z\alpha\alpha\beta} \\ &+83160 \cdot R_z^8 \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\beta\beta} \\ &-2835 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\beta\beta\delta\delta} \\ &-5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} \\ &-5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\beta} \cdot \Phi_{\alpha\beta\zeta\zeta} \\ &-5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} \\ &-5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\gamma} \cdot \Phi_{\alpha\gamma\epsilon\epsilon} \\ &-5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\gamma} \cdot \Phi_{\alpha\gamma\zeta\zeta} \\ &-17010 \cdot R_z^8 \cdot \Phi_{zz\alpha\delta} \cdot \Phi_{\alpha\delta\epsilon\epsilon} \\ &-5670 \cdot R_z^8 \cdot \Phi_{zz\alpha\delta} \cdot \Phi_{\alpha\delta\zeta\zeta} \\ &-5670 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\ &-5670 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\ &-5670 \cdot R_z^8 \cdot \Phi_{z\alpha\delta\delta} \cdot \Phi_{z\alpha\epsilon\epsilon} \\ &-22680 \cdot R_z^8 \cdot \Phi_{z\alpha\epsilon\zeta} \cdot \Phi_{z\alpha\epsilon\zeta} \\ &-5670 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\ &-5670 \cdot R_z^8 \cdot \Phi_{z\alpha\alpha\delta} \cdot \Phi_{z\delta\zeta\zeta} \\ &-5670 \cdot R_z^8 \cdot \Phi_{z\alpha\delta\delta} \cdot \Phi_{z\alpha\zeta\zeta} \end{aligned} \right] \quad (7)$$

Combining everything

$$\begin{aligned}
 T_{\alpha\beta\gamma\delta\epsilon\zeta\eta\theta}\Phi_{\alpha\beta\gamma\delta}\Phi_{\epsilon\zeta\eta\theta} = & \left[\begin{aligned}
 & +4 \cdot R_z^{-9} \cdot \Phi_{zzz\delta} \cdot \Phi_{zzz\delta} \\
 & +4 \cdot R_z^{-9} \cdot \Phi_{zzz\gamma} \cdot \Phi_{zzz\gamma} \\
 & +4 \cdot R_z^{-9} \cdot \Phi_{zzz\beta} \cdot \Phi_{zzz\beta} \\
 & +4 \cdot R_z^{-9} \cdot \Phi_{zzz\alpha} \cdot \Phi_{zzz\alpha} \\
 & -2/13 \cdot R_z^{-9} \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\eta\eta} \\
 & -4/13 \cdot R_z^{-9} \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\zeta\zeta} \\
 & -6/13 \cdot R_z^{-9} \cdot \Phi_{zzz\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\
 & -2/13 \cdot R_z^{-9} \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\eta\eta} \\
 & -4/13 \cdot R_z^{-9} \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
 & -6/13 \cdot R_z^{-9} \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\
 & -1/13 \cdot R_z^{-9} \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\eta\eta} \\
 & -12/13 \cdot R_z^{-9} \cdot \Phi_{zz\gamma\delta} \cdot \Phi_{zz\gamma\delta} \\
 & -2/13 \cdot R_z^{-9} \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\zeta\zeta} \\
 & -3/13 \cdot R_z^{-9} \cdot \Phi_{zz\gamma\gamma} \cdot \Phi_{zz\epsilon\epsilon} \\
 & -2/13 \cdot R_z^{-9} \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\eta\eta} \\
 & -4/13 \cdot R_z^{-9} \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\zeta\zeta} \\
 & -6/13 \cdot R_z^{-9} \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\epsilon\epsilon} \\
 & -2/13 \cdot R_z^{-9} \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\eta\eta} \\
 & -12/13 \cdot R_z^{-9} \cdot \Phi_{zz\beta\delta} \cdot \Phi_{zz\beta\delta} \\
 & -4/13 \cdot R_z^{-9} \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\zeta\zeta} \\
 & -6/13 \cdot R_z^{-9} \cdot \Phi_{zz\beta\beta} \cdot \Phi_{zz\epsilon\epsilon} \\
 & -12/13 \cdot R_z^{-9} \cdot \Phi_{zz\beta\gamma} \cdot \Phi_{zz\beta\gamma} \\
 & -4/13 \cdot R_z^{-9} \cdot \Phi_{zzz\delta} \cdot \Phi_{z\beta\beta\delta} \\
 & -4/13 \cdot R_z^{-9} \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\beta\beta\gamma} \\
 & -4/13 \cdot R_z^{-9} \cdot \Phi_{zzz\beta} \cdot \Phi_{z\beta\gamma\gamma} \\
 & -2/13 \cdot R_z^{-9} \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\eta\eta} \\
 & -4/13 \cdot R_z^{-9} \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\zeta\zeta} \\
 & -6/13 \cdot R_z^{-9} \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\epsilon\epsilon} \\
 & -3/13 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\eta\eta} \\
 & -12/13 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\delta} \cdot \Phi_{zz\alpha\delta} \\
 & -6/13 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\zeta\zeta} \\
 & -9/13 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{zz\epsilon\epsilon} \\
 & -12/13 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\gamma} \cdot \Phi_{zz\alpha\gamma} \\
 & -8/13 \cdot R_z^{-9} \cdot \Phi_{zzz\delta} \cdot \Phi_{z\alpha\alpha\delta} \\
 & -8/13 \cdot R_z^{-9} \cdot \Phi_{zzz\gamma} \cdot \Phi_{z\alpha\alpha\gamma} \\
 & -4/13 \cdot R_z^{-9} \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\gamma\gamma} \\
 & -12/13 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\beta} \cdot \Phi_{zz\alpha\beta} \\
 & -8/13 \cdot R_z^{-9} \cdot \Phi_{zzz\beta} \cdot \Phi_{z\alpha\alpha\beta} \\
 & -8/13 \cdot R_z^{-9} \cdot \Phi_{zzz\alpha} \cdot \Phi_{z\alpha\beta\beta} \\
 & +3/143 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\beta\beta\delta\delta} \\
 & +6/143 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\beta} \cdot \Phi_{\alpha\beta\epsilon\epsilon} \\
 & +6/143 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\beta} \cdot \Phi_{\alpha\beta\zeta\zeta} \\
 & +6/143 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\alpha} \cdot \Phi_{\gamma\gamma\delta\delta} \\
 & +6/143 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\gamma} \cdot \Phi_{\alpha\gamma\epsilon\epsilon} \\
 & +6/143 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\gamma} \cdot \Phi_{\alpha\gamma\zeta\zeta} \\
 & +18/143 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\delta} \cdot \Phi_{\alpha\delta\epsilon\epsilon} \\
 & +6/143 \cdot R_z^{-9} \cdot \Phi_{zz\alpha\delta} \cdot \Phi_{\alpha\delta\zeta\zeta} \\
 & +6/143 \cdot R_z^{-9} \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\gamma\epsilon\epsilon} \\
 & +6/143 \cdot R_z^{-9} \cdot \Phi_{z\alpha\alpha\delta} \cdot \Phi_{z\delta\epsilon\epsilon} \\
 & +6/143 \cdot R_z^{-9} \cdot \Phi_{z\alpha\delta\delta} \cdot \Phi_{z\alpha\epsilon\epsilon} \\
 & +24/143 \cdot R_z^{-9} \cdot \Phi_{z\alpha\epsilon\zeta} \cdot \Phi_{z\alpha\epsilon\zeta} \\
 & +6/143 \cdot R_z^{-9} \cdot \Phi_{z\alpha\alpha\gamma} \cdot \Phi_{z\gamma\zeta\zeta} \\
 & +6/143 \cdot R_z^{-9} \cdot \Phi_{z\alpha\alpha\delta} \cdot \Phi_{z\delta\zeta\zeta} \\
 & +6/143 \cdot R_z^{-9} \cdot \Phi_{z\alpha\delta\delta} \cdot \Phi_{z\alpha\zeta\zeta}
 \end{aligned} \right]
 \end{aligned} \tag{8}$$